TEAM BEYOND EARTH · 813 CHALLENGE

Harvest or stress

Alfalfa in Wadi as-Sirhan is cut every thirty days. A mown field and a dying field look the same from orbit — so the obvious way to rank fields for irrigation sends water to the fields that need it least. We measured how often, and built the fix.

Proof of Concept

Al Jouf, Saudi Arabia

February – September 2026

Precision Agriculture & Crop Intelligence · Sentinel-2 and NASA EMIT, open data only

THE PLACE

Wadi as-Sirhan runs on water that does not come back

Fossil groundwater

The aquifer under Al Jouf does not recharge. Every litre pumped is gone. Water, not land, is what limits this landscape.

Centre pivots, at scale

We detected **518 irrigated circles** covering **24,038 hectares** in a single 67 by 61 kilometre box. The belt extends well past it.

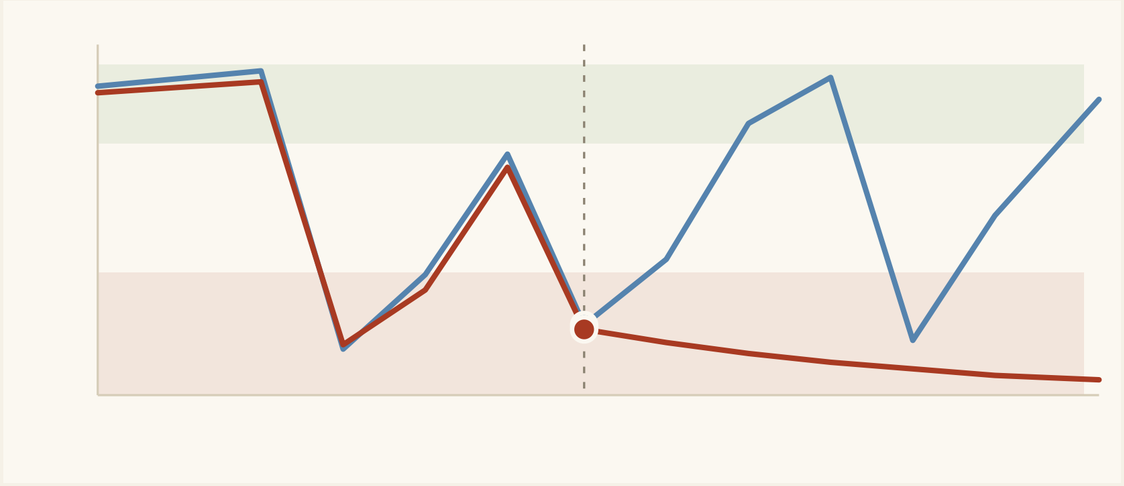
Fodder, cut monthly

The crop is alfalfa, harvested roughly every thirty days, all year. That cutting rhythm is the whole problem.

Area of interest 38.15, 29.75, 38.85, 30.30 · EPSG:32637 · field count and area measured, not estimated

THE CONFOUND

Vegetation here is a sawtooth, not a trend



Every cut drops a field from about **0.90** NDVI to about **0.35** within ten days. Then it regrows, and thirty days later it is cut again.

On the dashed line both fields read the same. Only what happens *afterwards* tells them apart.

Blue — cut, regrows on schedule

Red — stressed, never comes back

Schematic of the mechanism. NDVI levels and the ~30-day cadence are taken from the measured series.

THE MEASUREMENT · 8 SEPTEMBER 2026, THE SEASON’S LAST DATE

16 of the twenty fields a decline ranking sends you to first had simply been harvested

Between **seven and twenty** on every date from mid-April onward, and ten or more on 56 of those 61 dates. This is not a bad day — it is the normal behaviour of a decline ranking on a crop that gets cut. That ranking and ours agree on **one field out of twenty**.

Date	28 Apr	18 May	7 Jun	25 Jun	12 Jul	30 Jul	18 Aug	7 Sep
Of its top 20, simply mown	11	17	13	11	12	11	15	15

518 detected pivots, 82 cloud-screened days. Status is set by each field's own cutting rhythm, not by a threshold. Full per-date sweep: results/comparison_naive_vs_cycle.csv in the repository.

THE TOOL

Judge every field against its own rhythm, not against its neighbours

01

Find the cuts

A peak counts only if it stands 0.20 NDVI above the low points on both sides. That prominence test separates a real cut from a cloudy pixel.

02

Measure its cycle

Median spacing between that field's peaks, bounded to 18–60 days. Fewer than two clean peaks and we say so rather than invent one.

03

Expect a value

What did *this field* read this many days after each of its previous cuts? A low-vigour field is judged against itself.

04

Rank the deficit

Priority = how far behind its own history, times hectares. Exactly zero when a field is on schedule.

Because priority is zero for a field that is on schedule, **a freshly cut field can never enter the queue.** That is the entire design.

Runs on free Sentinel-2. No hyperspectral data required.

WHAT THE OPERATOR READS

Every field gets a sentence a farm manager can act on

ON CYCLE · PRIORITY 0

“Cut 45 days ago on a 45-day cycle. The drop is on schedule — this is harvest, not stress.”

Never enters the inspection queue.

OVERDUE · INSPECT FIRST

“Cut 47 days ago on a 32-day cycle and still at NDVI 0.19. It should have regrown by now — inspect.”

Ranked by deficit × hectares.

OUT OF ROTATION · PRIORITY 0

“Down for 138 days — more than three of its own 22-day cycles — with no regrowth. This field is out of rotation, not late.”

228 of 518 fields at season end. Adding this status cut the queue from 273 fields to 45.

All three read as a low NDVI in a single image. The difference is entirely in what each field's own history says should have happened by now — and a field that has been out of production all summer is not an irrigation problem at all.

Wording generated by the tool, not written for this slide.

WHAT WE PROCESSED

Field boundaries are derived, not drawn

518	24,038	82	33,078	30.5
pivots detected	hectares	cloud-screened days	field-days of NDVI	day cycle, median

Detection is a shape test

NDVI above 0.30, then keep only what is shaped like a pivot: 10–400 ha, eccentricity at most 0.75, solidity at least 0.85. No boundaries were supplied and none were drawn by hand.

Detect at peak green

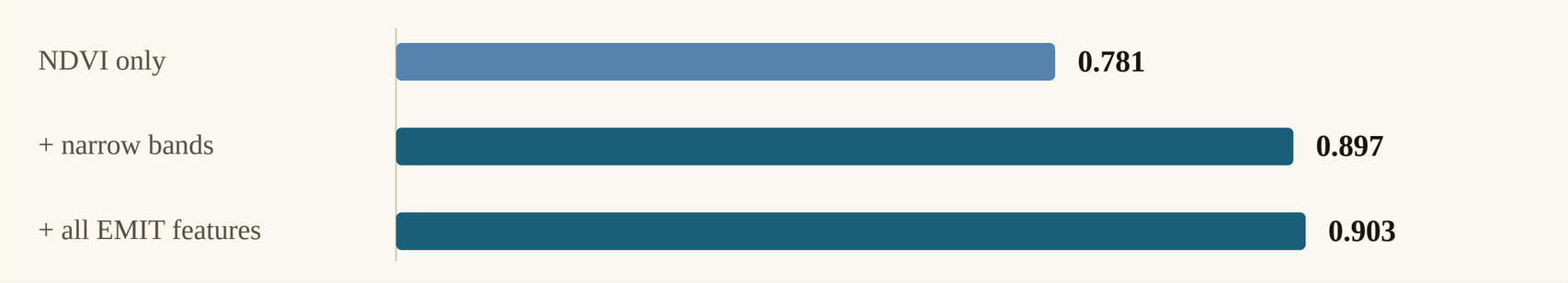
The date is chosen automatically: the greenest fully covered day of the season. Here, **13 April** — p95 NDVI 0.887, 100% clear. Eight of fourteen probed dates were discarded for partial coverage. On an earlier run, detecting in April found 511 fields and detecting in July only 328 — the same confound, attacking the step that builds the field set.

Sentinel-2 L2A via Microsoft Planetary Computer, cloud < 20%, mosaicked by solar day, SCL classes 0/1/3/8/9/10/11 removed. 20 m for detection, 60 m for the series.

THE HYPERSPECTRAL QUESTION

Our own best idea, set up as a fair test

Sentinel-2 has no band at 531, 570 or 970 nm. NASA's EMIT has 285 of them. The claim was that narrow bands could separate a cut field from a stressed one where NDVI cannot. We tested it as a prediction contest: labels taken from what each field did **15–45 days afterwards**, rows restricted to fields already low on the day, folds split by field, twenty repeats.



Better in **20 of 20 repeats**, and it survives removing EMIT's own NDVI — so it is not merely a second NDVI measurement. On this evidence the differentiator is proven.

286 field-observations · 22 recover, 264 stay low · cross-validated AUC, mean of 20 repeats · from the run with NASA Earthdata credentials supplied

THEN WE CHECKED IT PROPERLY

Only two EMIT scenes cleared the matching rule — so we held one out

Trained on one scene, tested on the other	n	recover	NDVI	+ narrow	+ all
Tested on 18 February	29	3	**0.962**	0.372	0.436
Tested on 22 July	257	19	**0.793**	0.629	0.655

The +0.12 was the model recognising which scene a row came from.

A February acquisition does not look like a July one. With two scenes, random folds cannot separate plant physiology from atmospheric conditions.

Adding the narrow bands makes prediction **worse** on a scene the model has not seen — in one case worse than a coin flip.

2 of 60 EMIT granules over this area matched a Sentinel-2 day within two days. Leave-one-scene-out needs six to eight.

WHAT WE SHIP

Three things, each one usable on its own

Dashboard

Map of every detected pivot, the ranked inspection queue, each field's own NDVI history with its cuts marked, and a side-by-side against the naive decline ranking.

API

A read-only service — **/priority**, **/fields/{id}**, **/comparison** — so the ranking feeds a farm's existing software instead of demanding a new system.

Notebook

One notebook rebuilds every number in the submission from open archives. A verification suite runs it end to end offline and holds the API and the dashboard to identical output.

No proprietary data, no supplier who can change terms, no credentials in the repository. Sentinel-2 and EMIT are openly licensed and the analysis code is ours.

The API is the integration path: most farms already have software; none of it models the cutting cycle.

THE CASE

Cost scales with area, not with customers

Who buys

Pivot operators — scouting time is the scarce resource.
Water regulators — allocation needs per-field evidence.
Lenders and insurers — crop condition without site visits.
Farm software — they own the customer, we own the cycle model.

Priced per hectare per season

We have deliberately not set the rate. It depends on what one avoided inspection is worth to a real operator, and that is the first question a pilot answers.

The unit economics

One area-season — 518 fields, 82 dates, 33,078 field-days — runs end to end on a free-tier notebook, on imagery that costs nothing and needs no credentials.

The marginal cost of adding a field is effectively zero.

What is defensible

Not the maths — anyone can compute NDVI. It is having tested the obvious approach, found it wrong, and published the check that overturned our own headline.

Field counts and dataset size measured from the committed run. Pricing shape is a stated modelling choice, not researched market data.

HONEST LIMITS, AND WHAT COMES NEXT

What this is not, and what we would do about it

Not yet validated

No field visit, no irrigation log, no yield record. Every status is a screening result. The tool says “look at this field first”, never “this field is water-stressed”.

Needs two cuts

A field with fewer than two clean peaks gets the regional 30-day default and is flagged as such. Newly planted fields are the weak case.

Relative, not absolute

A field declining across several cycles is judged against its own already-degraded history, and will under-flag.

If we go to incubation

1 · Ground truth. One operator, one season of irrigation records against our queue. Turns a screening tool into a measured one and sets the price.

2 · Regional scale-up. The full Wadi as-Sirhan belt and one UAE region. A compute cost, not a research cost.

3 · Deploy the API. Auth, rate limits, published spec, one reference integration.

4 · More scenes. Six to eight matched EMIT acquisitions would settle the question we left open. Upside, not dependency.

Every limitation above is in the methodology report, with the measurement that establishes it.

Sending water to a field that was mown last week is not a rounding error in an aquifer that does not refill.

We measured the mistake, built the tool that avoids it, and tested our own best idea hard enough to find out it was wrong. All of it on open data, all of it reproducible.

Team Beyond Earth

Arab Youth Space Hackathon 2026

813 Challenge · Theme 02

Copernicus Sentinel-2 · NASA EMIT · both open. Analysis code our own.

SPEAKER NOTES · SLIDES 1–3

1 Harvest or stress

Open on the tension in the title. The whole submission turns on one measured fact: a decline ranking here ranks harvest, not stress. Everything after this slide is evidence for that and the tool that fixes it.

2 Wadi as-Sirhan runs on water that does not come back

Set the stake before the method. The reason this matters is that the input being misallocated is irreplaceable. Field count and hectares are from our own detection, reproducible from the notebook.

3 Vegetation here is a sawtooth, not a trend

Say plainly this chart is a schematic, not measured data — the measured version is the next slide. The point to land: rank by decline and you are ranking by how recently the mower came through.

4

16

This is the number the whole submission rests on, and it is measured on all 69 dates with enough history to model, not on one date. The point to land: the failure is not occasional. On 22 July specifically, 57 of the 518 fields were mid-cut. If pressed on the lower bound: the single clean date all season is 27 March, before the cutting rhythm has established itself.

5

Judge every field against its own rhythm, not against its neighbours

Four steps, and the fourth is the one that matters. Land the last line: the tool is built so that the failure mode on slide four is structurally impossible, not merely less likely.

6

Every field gets a sentence a farm manager can act on

Show that the output is operational language, not an index value. All three quotes are copied verbatim from results/priority_queue_2026-09-08.csv — what the tool actually wrote about fields 201, 392 and 277. Dwell on the third: we only found the out-of-rotation case because the first queue was implausibly large at 273 fields, and fixing it removed 228 fields that were never candidates.

SPEAKER NOTES · SLIDES 7–9

7 Field boundaries are derived, not drawn

The 511-versus-328 difference is worth pausing on: it is the same confound appearing in the detection step, and it is why the detection date is chosen by the algorithm rather than by us. The cycle length is measured, not assumed — a median of 30.5 days across 493 of the 518 fields; the other 25 never showed a clean peak and fall back to a 30-day default.

8 Our own best idea, set up as a fair test

Deliver this slide straight, as if the finding stands. The next slide is the turn. Do not hedge here — the contrast is the point.

9 Only two EMIT scenes cleared the matching rule — so we held one out

This is the slide that should win technical credibility. We found the artefact in our own headline result before shipping it. The check took four lines of code and is the difference between a finding and an artefact. Say that we report the narrow bands as measurement and rank nothing with them.

SPEAKER NOTES · SLIDES 10–12

10 Three things, each one usable on its own

Emphasise deployability — this is the criterion that rewards a working interface over four matplotlib panels. The API framing matters commercially: we do not have to displace an incumbent to be useful.

11 Cost scales with area, not with customers

Be explicit that we did not invent a price. Judges on criterion 7 respond better to a model with named assumptions than to a fabricated TAM. Every figure on this slide comes from our own committed run.

12 What this is not, and what we would do about it

Do not soften these. The limitation slide is where a technical jury decides whether to trust the rest of the deck. The strongest thing we can say is that we know exactly what we have not proven.

13

Sending water to a field that was mown last week is not a rounding error in an aquifer that does not refill.

Close on the stake, not on a thank-you slide. Then take questions; the limitations slide is the one to return to if the jury probes.